

# HONGYU HÈ

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( Version: July 2026 | 📄 indicates a hyperlink )

## EDUCATION

|                                                                                                                  |                       |
|------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Princeton University — PhD in Electrical Engineering</b>                                                      | Aug. 2024 – present   |
| ▶ Committee: Maria Apostolaki <sup>📄</sup> (advisor), Sharad Malik <sup>📄</sup> , Aarti Gupta <sup>📄</sup>       | United States         |
| ▶ GPA: 3.95/4.0 (8 graduate-level courses in EE and CS)                                                          |                       |
| <b>ETH Zürich — MSc in Computer Science</b>                                                                      | Aug. 2021 – Jul. 2024 |
| ▶ Direct Doctoral Scholarship <sup>📄</sup> (5 out of 230 CS master's students in 2022)                           | Switzerland           |
| ▶ Thesis awarded highest grade: 6.0/6.0 (Advisor: Gustavo Alonso <sup>📄</sup> )                                  |                       |
| ▶ Associated Researcher <sup>📄</sup> at ETH AI Center <sup>📄</sup>                                               |                       |
| <b>Vrije Universiteit Amsterdam — BSc (Hons) in Computer Science</b>                                             | Aug. 2018 – Jul. 2021 |
| <b>Universiteit van Amsterdam — BSc Minor in Mathematics</b>                                                     | Sep. 2019 – Jan. 2021 |
| ▶ GPA rank: 1st/180 <sup>📄</sup> ; national top 8 in CS <sup>📄</sup> (KHMW <sup>📄</sup> 2020)                    | The Netherlands       |
| <b>Tianjin University — BA in Law</b> (unfinished)                                                               | Sep. 2016 – Aug. 2017 |
| ▶ Top 0.2% among 153,276 regional liberal arts students in National College Entrance Exam, “Gaokao” <sup>📄</sup> | China                 |

## INDUSTRY EXPERIENCE\*

|                                                                                                                  |                       |
|------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Microsoft<sup>📄</sup> — Research Intern</b>   Networking Research Group                                       | Jun. 2026 – Sep. 2026 |
| <b>Curieo AI<sup>📄</sup> (sold) — ML Research Fellow</b>   remote, part-time                                     | Jul. 2024 – Oct. 2024 |
| <b>IBM Research<sup>📄</sup> — Engineering Intern</b>   Geodata Team   part-time                                  | Apr. 2024 – Jun. 2024 |
| <b>Apple Inc.<sup>📄</sup> — AI/ML Intern</b>   Visual Intelligence Team   (return offer)                         | May 2023 – Oct. 2023  |
| <b>Oracle Labs<sup>📄</sup> — Graal Cloud Native Intern</b>   Serverless Team   part-time   (return-intern offer) | Feb. 2023 – May 2023  |
| <b>Dexter Energy Services<sup>📄</sup> — BSc Thesis Intern</b>                                                    | Jun. 2021 – Sep. 2021 |
| <b>Huawei Technologies<sup>📄</sup> (Amsterdam Research Center) — R&amp;D Intern</b>   Search Team                | Mar. 2021 – Jun. 2021 |
| <b>Picnic Technologies<sup>📄</sup> — Software Developer</b>   Store Team   part-time                             | Sep. 2020 – Mar. 2021 |
| — Backend Engineer Intern   Mentor: Sander Mak <sup>📄</sup>   (return offer)                                     | Jun. 2020 – Sep. 2020 |

## TEACHING

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|---------------------------------------------------------------------------------------------------------|------------|
| <b>Princeton University</b>   20 hrs/wk (head TA)   Computer Networks <sup>📄</sup>                      | 2025       |
| <b>ETH Zürich</b>   14 hrs/wk   Distributed Systems Lab, Big Data (my session recordings <sup>📄</sup> ) | 2023, 2024 |
| <b>Vrije Universiteit Amsterdam</b>   8–16 hrs/wk   Graph Theory, Linear Algebra, C++, Calculus I       | 2019, 2020 |

## TECHNICAL SKILLS

**Programming & Scripting:** C/C++, Python, (System)Verilog, Golang, Shell, Scala, Java, SQL,  $\LaTeX$   
**Software & Tools:** Kubernetes, Xilinx Vivado/Vitis suite, JAX, PyTorch, Linux, Git, Docker, Knative, Apache Spark, SLURM, STM32 (ARM Cortex-M3), MongoDB, PostgreSQL, Hadoop, HBase, Spring5, Qt5

\*Industry recommendations: hhy.ee.princeton.edu/rec/<sup>📄</sup>

## PUBLICATIONS

### Preprint

- [12] H. Hè, M. Apostolaki. *Let AI Agents Translate Networks, Not Reason About Them*. [PDF<sup>□</sup>]
- [11] H. Hè<sup>\*</sup>, A. Krentsel<sup>\*</sup>, S. Ratnasamy, M. Apostolaki. *Invariant Discovery for Networked Systems*. [PDF<sup>□</sup>]
- [10] M. Jin, H. Hè, M. Apostolaki. *Cross-Flow Correlations Survive Synthesis: Measuring Source-Level Privacy Leakage in Synthetic Network Traces*. [PDF<sup>□</sup>]

### Peer-reviewed paper

- [9] H. Hè, M. Jin, M. Apostolaki. *Worst-Case Discovery and Runtime Protection for RL-Based Network Controllers*. USENIX NSDI 2027 (to appear) [PDF<sup>□</sup>]
- [8] H. Hè, M. Jin, M. Apostolaki. *Making Logic a First-Class Citizen in Generative ML for Networking*. USENIX NSDI 2026 [code<sup>□</sup>] [PDF<sup>□</sup>]
- [7] H. Hè, M. Apostolaki. *Just-in-Time Logic Enforcement: A new paradigm of combining statistical and symbolic reasoning for network management*. ACM HotNets 2025 [code<sup>□</sup>] [PDF<sup>□</sup>]
- [6] A. Zhou, C. Costic, H. Hè, A. Ghalayini, A. Kabbani, M. Apostolaki. *Mitigating Inter-datacenter Incast with a Proxy: The shortest path is not necessarily the fastest*. ACM HotNets 2025 [PDF<sup>□</sup>]
- [5] B. McKinzie, Z. Gan, J. Fauconnier, ..., H. Hè, ..., P. Gräsch, A. Toshev, Y. Yang. *MMI: Methods, Analysis & Insights from Multimodal LLM Pre-training*. ECCV 2024 **(first flagship multimodal LLM from 🍏)** [PDF<sup>□</sup>]
- [4] H. Hè, M. Friedman, T. Rekatsinas. *EnergAt: Fine-Grained Energy Attribution for Multi-Tenancy*. ACM HotCarbon 2023 **(adopted by Trycarbonara<sup>□</sup>)** [media<sup>□</sup>] [code<sup>□</sup>] [PDF<sup>□</sup>]
- [3] B. Kruit, H. He, J. Urbani. *Tab2Know: Building a Knowledge Base from Tables in Scientific Papers*. ISWC 2020. [code<sup>□</sup>] [PDF<sup>□</sup>]

### Degree thesis

- [2] H. Hè. *FPGA-based SmartNIC for Distributed Machine Learning*. 2024 **(best grade: 6.0/6.0)** [MSc thesis<sup>□</sup>]
- [1] H. He. *How Can Datacenters Join the Smart Grid to Address the Climate Crisis? Using simulation to explore power and cost effects of direct participation in the energy market*. 2021 **(Amsterdam Thesis Award<sup>□</sup>)** [BSc thesis<sup>□</sup>]

## OTHER ACHIEVEMENTS

- ◆ Nominee for Princeton Teaching Award (3 nominees/department annually, see letters from my students here<sup>□</sup>) 2026
- ◆ Princeton Natural & Artificial Mind Fellowship in AI 2025
- ◆ Princeton University Administrative Fellowship in CS 2024
- ◆ Nominee for ETH Medal<sup>□</sup> (MSc) 2023
- ◆ Danish Government scholarship<sup>□</sup> [€ 28,260] 2021
- ◆ Leiden Excellence scholarship<sup>□</sup> [€ 15,000] 2021
- ◆ GRE 334 (verbal: 161, quant: 170, writing: 3) 2021
- ◆ GLOBE scholarship (ranked 1st out of 900+<sup>□</sup>) [€ 1,250] 2020
- ◆ ACM-ICPC: Amsterdam Algorithm Programming Preliminaries (AAPP) with my team, ranked 1st 2020
- ◆ ACM-ICPC: Benelux Algorithm Programming Contest (BAPC) with my team, ranked 9th 2019
- ◆ Young Talent Award<sup>□</sup> by *The Royal Holland Society of Sciences (KHMW)*<sup>□</sup> 2019
- ◆ Ranked 1st in Operating Systems (Cristiano Giuffrida<sup>□</sup>) and Networks (Andrew Tanenbaum<sup>□</sup>) courses 2019

## SERVICE

- **Artifact Evaluation Committee:** SOSP '23; MLSys '23; MobiSys '23; JSys '22, '23; ICPE '24
- **Tutorial Organizer:** ASPLOS '22 (Integration of Firecracker  $\mu$ VM<sup>□</sup>)

## LANGUAGES

**English:** TOEFL (speaking): 28, IELTS: 9 | **German:** B1 | **Dutch:** Basic | **Mandarin:** PSC<sup>□</sup>: Level 1-B, regional max.